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Project Title:	AD ASTRA (Anaerobic Digestion for Advanced Space Technology and Resource Acquisition)		
Division Name:	Space Biology		
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Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2024 NNH24ZDA001N-CIBS E.11 Consortium in Biological Sciences
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Pakrasi, Himadri (Washington University in St. Louis) Roberson, Luke (NASA Kennedy Space Center) Blenner, Mark (University of Delaware) Menezes, Amor (University of Florida) Averesch, Nils (University of Florida) Huang, Laibin (St. Louis University) Hurisso, Tunsisa (Lincoln University of Missouri)		
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Task Description:	We propose to establish a new consortium named AD ASTRA (Anaerobic Digestion for Advanced Space Technology and Resource Acquisition), consisting of five diverse universities (three R1, one R2, and one Historically Black Colleges and Universities/HBCUs) and a NASA Space Center. The AD ASTRA consortium will develop integrative biotechnology with proven performance in simulated microgravity and low gravity conditions to sustain astronauts on space missions. Our technology will apply a new anaerobic digestion (AD) process that converts human wastes into organic acids and carbon dioxide (CO2). This AD-derived carbon feedstock will be utilized for downstream biomanufacturing applications. Specifically, our team will engineer photosynthetic cyanobacterial monoculture and cocultures with heterotrophic microbial hosts via synthetic biology approaches to produce nutraceuticals, food, and biopolymers. Our focus on coculture performance exemplifies our integrative approach to reducing the mass, volume, and power costs that are associated with sequential space biomanufacturing modules and representing a substantive advance over the current state-of-the-art monoculture space synthetic biology. We will also validate and optimize our integrative AD and photo-bio-refinery in light of nearer-term lunar surface missions via the following aims: (1) examine microbial community and algal responses to simulated low gravity, emphasizing that of the Moon; (2) incorporate lunar regolith simulants and AD-derived wastewater as an alternate culture media; (3) integrate and optimize the biological systems from AD to photo-biorefinery via design of experiments, process modeling, and machine learning, and (4) perform experimental tests, systems biology analyses, and metabolic and kinetic modeling to understand microbial physiologies and population interactions under tested space conditions. The deep foundational new knowledge obtained will further our efforts to improve AD ASTRA applications via a Design-Build-Test-Learn cycle.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Space biomanufacturing, waste management, and in situ resource utilization can sustain the development of human habitats on the Moon, Mars, or on space stations in between. The proposed technology and engineering protocols reduce the launch mass and power usage on long-duration manned missions, and also create beneficial operating environments for astronauts. This work will generate a sustainable closed-loop space life support system by recovering resources from the waste stream and using them for microbial biomanufacturing. A well-established Design-Build-Test-Learn (D-B-T-L) cycle will result in tangible biotechnologies and new knowledge that can be applied on the lunar surface. These will increase mission-capabilities through extended life support, and they will be valuable for food and nutraceuticals to sustain offplanet operations.
Task Progress:	This reporting period focused on advancing integrated biological systems that support sustainable life support and in situ biomanufacturing for long-duration space missions. Across several collaborating institutions, the project addressed key challenges in wastewater recycling, nutrient recovery, carbon utilization, and production of high-value biomolecules under space-relevant conditions. The overall goal is to reduce dependence on Earth resupply by developing closed-loop microbial systems capable of converting waste streams into useful resources, including clean water, oxygen, nutritional supplements, and bioproducts. A major focus of the work was the development of synthetic microbial communities that combine photosynthetic and heterotrophic organisms into cooperative systems. Researchers at Washington University in St. Louis developed a hydrogel-based “synthetic lichen” composed of engineered cyanobacteria and yeast immobilized together in alginate beads. In this system, cyanobacteria convert carbon dioxide into sucrose through photosynthesis, while engineered yeast consumes the sucrose to produce β-carotene, a precursor to vitamin A and an important antioxidant relevant to astronaut health. The hydrogel platform was designed to function under resource-limited conditions while also supporting wastewater treatment and nutrient recycling. The co-culture system was successfully established without the need for an external carbon source, demonstrating stable metabolic cooperation between the two organisms. Medium composition and environmental conditions were optimized to improve microbial growth and sucrose production. Salt stress studies showed that hydrogel encapsulation improved the resilience of cyanobacteria under elevated salinity conditions relevant to wastewater treatment systems. These findings suggest that immobilized co-culture platforms may provide a robust strategy for maintaining microbial productivity in harsh or resource-constrained space environments. In parallel, the research team developed a large-scale knowledge graph to analyze trends and emerging concepts in algae-integrated anaerobic digestion and wastewater treatment systems. Using more than 10,000 research abstracts, the study identified major themes related to nutrient recovery, algal bioprocessing, bio-electrochemical systems, and space applications. The analysis highlighted hydrogel immobilization as a promising strategy for improving biomass retention and nutrient removal efficiency, while also identifying opportunities for future research in integrated biological life support technologies. Additional studies explored the use of extremophile microorganisms for sustainable biomanufacturing. One promising organism, Colwellia psychrerythraea 34H, demonstrated stable growth and biopolymer production under low temperature, high salinity, and simulated microgravity conditions. The organism successfully produced polyhydroxybutyrate (PHB), a biodegradable polymer with potential applications in space manufacturing. Isotope labeling experiments confirmed that a significant portion of the polymer carbon originated from recycled carbon sources, supporting the feasibility of closed-loop carbon recycling systems for future missions. Another major effort focused on evaluating wastewater-derived growth media for cyanobacterial cultivation. Researchers investigated the growth of engineered Synechococcus elongatus in media designed to mimic treated spacecraft wastewater produced by anaerobic membrane bioreactors. While the wastewater-derived media supported cyanobacterial growth, nutrient limitations, particularly iron deficiency, reduced productivity compared to standard laboratory media. Supplementation with iron and nitrate significantly improved growth and photosynthetic activity, demonstrating that relatively small nutrient adjustments may enable productive cultivation using recycled waste streams. These findings support the use of anaerobic digestion technologies as a foundation for integrated resource recovery systems in space habitats. The project also expanded into development of multifunctional microbial consortia capable of producing protective biomaterials. A synthetic co-culture was established using cyanobacteria and engineered Vibrio natriegens for melanin production. In this system, photosynthetically produced sucrose from cyanobacteria served as the carbon source for melanin synthesis by the bacterial partner. Melanin is of particular interest for space applications because of its ability to absorb ultraviolet and ionizing radiation, offering potential uses in radiation shielding and protective biomaterials.

Optimization studies identified culture conditions that enhanced melanin production, and successful co-culture experiments demonstrated fully light-driven synthesis of the pigment. This work provides a model for future division-of-labor microbial systems in which individual organisms specialize in carbon fixation or targeted bioproduct formation.

To better understand microbial processes involved in waste treatment, researchers at Saint Louis University characterized microbial communities present in human waste samples relevant to anaerobic digestion systems. DNA extraction, quantitative PCR, and 16S rRNA sequencing were used to identify bacterial, archaeal, and methanogenic populations. Results revealed diverse microbial communities associated with nutrient cycling and methane production, including several methanogenic groups important for anaerobic digestion performance. These studies established baseline protocols and datasets that will support future monitoring and optimization of biological waste treatment systems for space applications.

At the University of Delaware, efforts focused on improving sucrose utilization in non-conventional yeast species used in synthetic co-culture systems. Engineered strains of Pichia pastoris and Yarrowia lipolytica were modified with sucrose transport and metabolism pathways to improve carbon assimilation and β-carotene production. The engineered strains demonstrated enhanced sucrose consumption and established a foundation for more stable and efficient carbon exchange between photosynthetic and heterotrophic organisms. Ongoing work aims to further improve intracellular sucrose uptake while minimizing extracellular sugar leakage that could destabilize co-culture performance.

Researchers at the University of Florida pursued complementary efforts in microbial polymer production and reduced-gravity bioprocessing. One project focused on engineering Cupriavidus necator for production of poly(lactic acid)-based biopolymers. Genetic modifications were introduced to prevent carbon loss through competing metabolic pathways and to improve polymer synthesis. Preliminary results indicated increased incorporation of lactic acid into the polymer structure, although additional optimization is needed to improve overall yield and material properties.

A second University of Florida effort examined how simulated microgravity affects anaerobic digestion systems and microbial co-cultures. Experimental platforms were established using rotating wall vessel bioreactors to study microbial growth and metabolic activity under low-gravity conditions. Initial studies demonstrated that engineered cyanobacteria, E. coli, and extremophile bacteria could maintain growth under simulated microgravity, although some reductions in growth rate were observed. Ongoing experiments are evaluating sucrose production, wastewater utilization, and co-culture stability under these conditions. In parallel, computational modeling tools were expanded to better predict biological system performance in reduced gravity and to support future design of autonomous space biomanufacturing systems.

Overall, the project made substantial progress toward development of integrated biological platforms capable of supporting sustainable human exploration beyond Earth. The work demonstrated advances in wastewater-derived microbial cultivation, hydrogel-based synthetic ecosystems, radiation-protective biomaterial production, microbial community characterization, polymer synthesis, and reduced-gravity bioprocessing. Together, these accomplishments strengthen the foundation for future closed-loop life support systems that combine waste recycling, carbon capture, nutrient recovery, and in situ manufacturing into unified biological platforms for long-duration space missions.

Bibliography Type:	Description: (Last Updated: 05/25/2026)
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